Лабораторная работа №1
Компьютерный практикум по статистическому анализу данных

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Установка необходимых приложений

```
PS C:\WINDOWS\system32> Set-ExecutionPolicy Bypass -Scope Process -Force; [System.Net.ServicePointManager]::SecurityProt
ocol = [System.Net.ServicePointManager]::SecurityProtocol -bor 3072; iex ((New-Object System.Net.WebClient).DownloadStri
ng('https://community.chocolatey.org/install.ps1'))
ПРЕДУПРЕЖДЕНИЕ: 'choco' was found at 'C:\ProgramData\chocolatey\bin\choco.exe'.
ПРЕДУПРЕЖДЕНИЕ: An existing Chocolatey installation was detected. Installation will not continue. This script will not
overwrite existing installations.
If there is no Chocolatey installation at 'C:\ProgramData\chocolatey', delete the folder and attempt the installation
again.

Please use choco upgrade chocolatey to handle upgrades of Chocolatey itself.
If the existing installation is not functional or a prior installation did not complete, follow these steps:
 - Backup the files at the path listed above so you can restore your previous installation if needed.
 - Remove the existing installation manually.
 - Rerun this installation script.
 - Reinstall any packages previously installed, if needed (refer to the lib folder in the backup).

Once installation is completed, the backup folder is no longer needed and can be deleted.
PS C:\WINDOWS\system32>
```

Рис. 1: Установка менеджера пакетов Chocolatey

Установка необходимых приложений

```
PS C:\WINDOWS\system32> choco install far -y
Chocolatey v2.4.1
Installing the following packages:
far
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading Far 3.0.6446... 100%

far v3.0.6446 [Approved]
far package files install completed. Performing other installation steps.
Downloading Far 64 bit
  from 'https://www.farmanager.com/files/Far30b6446.x64.20250301.msi'
Progress: 100% - Completed download of C:\Users\Пользователь\AppData\Local\Temp\chocolatey\Far\3.0.6446\Far30b6446.x64.2
0250301.msi (14.08 MB).
Download of Far30b6446.x64.20250301.msi (14.08 MB) completed.
Hashes match.
Installing Far...
Far has been installed.
  far may be able to be automatically uninstalled.
 The install of far was successful.
  Deployed to 'C:\Program Files\Far Manager\'

Chocolatey installed 1/1 packages.
 See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).
PS C:\WINDOWS\system32>
```

Рис. 2: Установка Far Manager

Установка необходимых приложений

```
PS C:\WINDOWS\system32> choco install notepadplusplus -y
Chocolatey v2.4.1
Installing the following packages:
notepadplusplus
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading notepadplusplus.install 8.8.8... 100%

notepadplusplus.install v8.8.8 [Approved]
notepadplusplus.install package files install completed. Performing other installation steps.
Installing 64-bit notepadplusplus.install...
notepadplusplus.install has been installed.
WARNING: No registry key found based on  'Notepad\+\+*'
notepadplusplus.install installed to 'C:\Program Files\Notepad++'
Added C:\ProgramData\chocolatey\bin\notepad++.exe shim pointed to 'c:\program files\notepad++\notepad++.exe'.
  notepadplusplus.install can be automatically uninstalled.
 The install of notepadplusplus.install was successful.
  Software installed as 'exe', install location is likely default.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading notepadplusplus 8.8.8... 100%

notepadplusplus v8.8.8 [Approved]
notepadplusplus package files install completed. Performing other installation steps.
 The install of notepadplusplus was successful.
  Software install location not explicitly set, it could be in package or
  default install location of installer.

Chocolatey installed 2/2 packages.
 See the log for details (C:\ProgramData\chocolatey\logs\chocolatey.log).
PS C:\WINDOWS\system32>
```

Рис. 3: Установка Notepad++

Установка необходимых приложений

```
PS C:\WINDOWS\system32> choco install julia
Chocolatey v2.4.1
Installing the following packages:
julia
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading Julia 1.12.0... 100%

julia v1.12.0 [Approved]
julia package files install completed. Performing other installation steps.
The package julia wants to run 'chocolateyinstall.ps1'.
Note: If you don't run this script, the installation will fail.
Note: To confirm automatically next time, use '-y' or consider:
choco feature enable -n allowGlobalConfirmation
Do you want to run the script?([Y]es/[A]ll - yes to all/[N]o/[P]rint): y

Installing 64-bit Julia...
```

Рис. 4: Установка Julia

Установка необходимых приложений

```
PS C:\WINDOWS\system32> choco install anaconda3 -y
Chocolatey v2.4.1
Installing the following packages:
anaconda3
By installing, you accept licenses for the packages.
Downloading package from source 'https://community.chocolatey.org/api/v2/'
Progress: Downloading anaconda3 2025.6.0... 100%

anaconda3 v2025.6.0 [Approved]
anaconda3 package files install completed. Performing other installation steps.
WARNING: The Anaconda3 installation can take a long time (up to 30 minutes).
WARNING: Please be patient and let it finish.
WARNING: If you want to verify the install is running, you can watch the installer process in Task Manager
Downloading anaconda3 64 bit
  from 'https://repo.anaconda.com/archive/Anaconda3-2025.06-0-Windows-x86_64.exe'
Progress: 100% - Completed download of C:\Users\Пользователь\AppData\Local\Temp\anaconda3\2025.6.0\Anaconda3-2025.06-0-W
indows-x86_64.exe (914.33 MB).
Download of Anaconda3-2025.06-0-Windows-x86_64.exe (914.33 MB) completed.
Hashes match.
Installing anaconda3...
```

Рис. 5: Установка Anaconda Distribution (Python 3.x)

Установка необходимых приложений

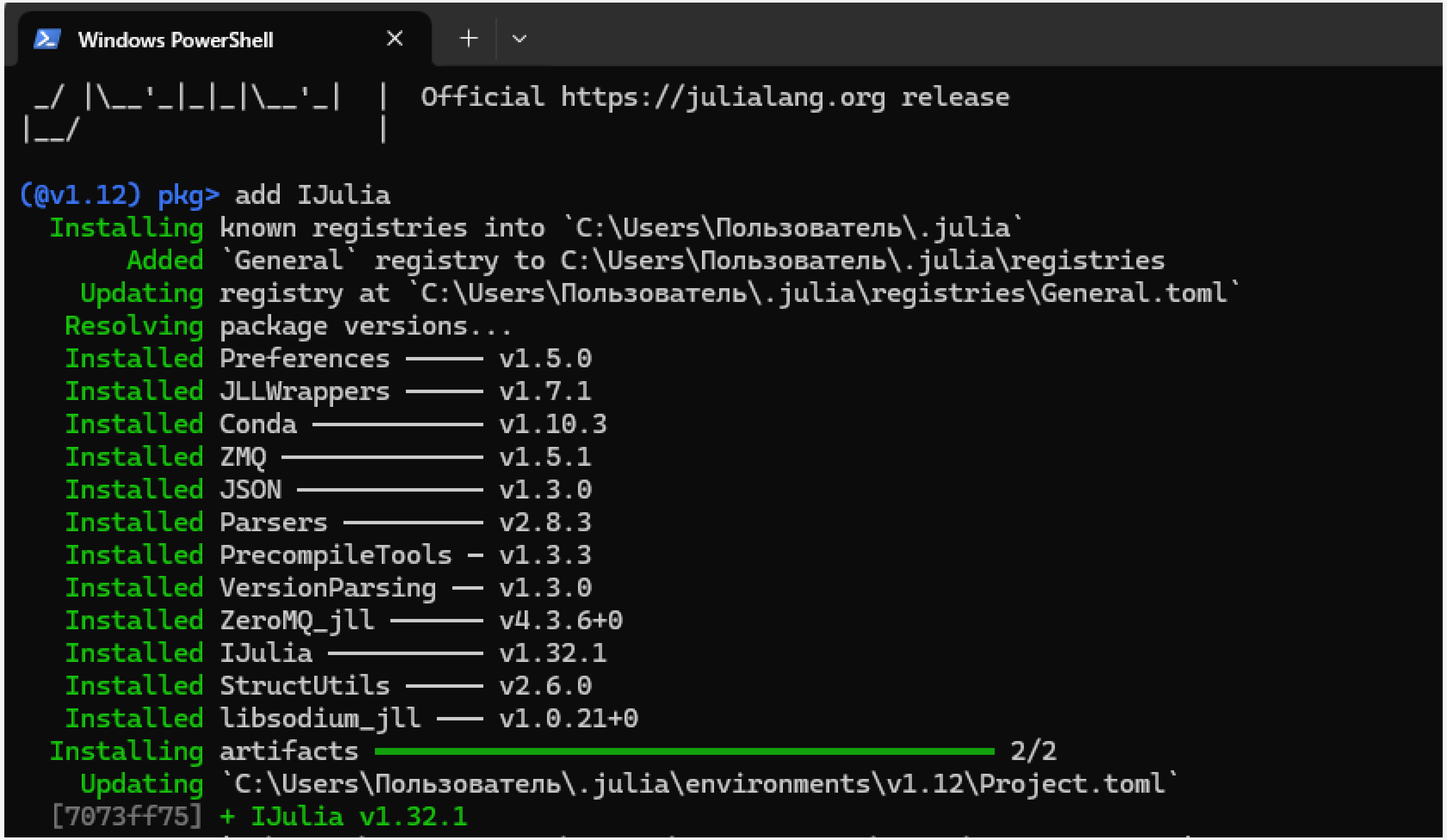

Рис. 6: Установка пакетов для работы с Jupyter

Примеры

[5]:
```julia
typeof(3), typeof(3.5), typeof(3/3.55), typeof(sqrt(3+4im)), typeof(pi)
```

[5]:
```
(Int64, Float64, Float64, ComplexF64, Irrational{:π})
```

[6]:
```julia
1.0/0.0, 1.0/(-0.0), 0.0/0.0
```

[6]:
```
(Inf, -Inf, NaN)
```

[7]:
```julia
typeof(1.0/0.0), typeof(1.0/-0.0), typeof(0.0/0.0)
```

[7]:
```
(Float64, Float64, Float64)
```

[9]:
```julia
for T in [Int8, Int16, Int32, Int64, Int128, UInt8, UInt16, UInt32, UInt64, UInt128]
    println("$(lpad(T,7)): [$(typemin(T)),$(typemax(T))]")
end
```

```
   Int8: [-128,127]
  Int16: [-32768,32767]
  Int32: [-2147483648,2147483647]
  Int64: [-9223372036854775808,9223372036854775807]
 Int128: [-170141183460469231731687303715884105728,170141183460469231731687303715884105727]
  UInt8: [0,255]
 UInt16: [0,65535]
 UInt32: [0,4294967295]
 UInt64: [0,18446744073709551615]
UInt128: [0,340282366920938463463374607431768211455]
```

Рис. 7: Примеры определения типа числовых величин

Примеры

```
[10]: Int64(2.0), Char(2), typeof(Char(2))

[10]: (2, '\x02', Char)

[11]: convert(Int64, 2.0), convert(Char, 2)

[11]: (2, '\x02')

[14]: typeof(promote(Int8(1), Float16(4.5), Float32(4.1)))

[14]: Tuple{Float32, Float32, Float32}
```

Рис. 8: Примеры приведения аргументов к одному типу

Примеры

```
[15]: function f(x)
          x^2
      end

[15]: f (generic function with 1 method)

[16]: f(4)

[16]: 16

[17]: g(x)=x^2

[17]: g (generic function with 1 method)

[18]: g(8)

[18]: 64
```

Рис. 9: Примеры определения функций

Примеры

```
[19]: a = [4 7 6] # вектор-строка
      b = [1, 2, 3] # вектор-столбец
      a[2], b[2] # вторые элементы векторов a и b

[19]: (7, 2)

[20]: a = 1; b = 2; c = 3; d = 4 # присвоение значений
      Am = [a b; c d] # матрица 2 x 2

[20]: 2×2 Matrix{Int64}:
       1  2
       3  4

[21]: Am[1,1], Am[1,2], Am[2,1], Am[2,2] # элементы матрицы

[21]: (1, 2, 3, 4)

[22]: aa = [1 2]
      AA = [1 2; 3 4]
      aa*AA*aa'

[22]: 1×1 Matrix{Int64}:
       27

[23]: aa, AA, aa'

[23]: ([1 2], [1 2; 3 4], [1; 2;;])
```

Рис. 10: Примеры работы с массивами

Примеры для задания №1

```
[24]: write("myfile.txt", "Hello World! \nMy name is Bulat")

[24]: 30

[26]: io = open("myfile.txt", "r")
      read(io, String)

[26]: "Hello World! \nMy name is Bulat"

[27]: readline("myfile.txt")

[27]: "Hello World! "

[28]: readlines("myfile.txt")

[28]: 2-element Vector{String}:
       "Hello World! "
       "My name is Bulat"

[29]: print("Hello World!")

      Hello World!

[31]: println("Hello", " World", "!")

      Hello World!

[32]: show("Hello World!")

      "Hello World!"
```

Рис. 11: Примеры работы с функциями для чтения/записи/вывода информации на экран

Примеры для задания №2

```
[33]:  ex1 = Meta.parse("(10 + 10) / 5")

[33]:  :((10 + 10) / 5)
```

Рис. 12: Пример работы с функцией parse

Примеры для задания №3

```
[34]: 1+1
[34]: 2
[35]: 5-3.0
[35]: 2.0
[36]: 5.0*3.0
[36]: 15.0
[37]: 8/2
[37]: 4.0
[38]: 9/5
[38]: 1.8
[39]: 9%5
[39]: 4
[40]: 10.0%3
[40]: 1.0
[41]: 3^3
[41]: 27
```

Рис. 13: Примеры работы базовых математических операций

Примеры для задания №3

```
[42]: 3 == 3

[42]: true

[43]: 3 != 3

[43]: false

[44]: 5.0 < 5

[44]: false

[45]: 5 >= 3

[45]: true

[46]: x = true
      !x

[46]: false

[47]: x = true
      y = false
      x || y

[47]: true
```

Рис. 14: Примеры работы базовых математических операций

Примеры для задания №4

```
[50]: using LinearAlgebra
      A = [1 3 5; 2 4 6; 7 9 8]

[50]: 3x3 Matrix{Int64}:
       1  3  5
       2  4  6
       7  9  8

[51]: det(A)

[51]: 6.0

[52]: tr(A)

[52]: 13

[53]: inv(A)

[53]: 3x3 Matrix{Float64}:
       -3.66667   3.5   -0.333333
        4.33333  -4.5    0.666667
       -1.66667   2.0   -0.333333

[54]: B = [1 1 1; 1 1 1; 1 1 1]
      A - B

[54]: 3x3 Matrix{Int64}:
       0  2  4
       1  3  5
       6  8  7

[55]: A + B

[55]: 3x3 Matrix{Int64}:
       2   4  6
       3   5  7
       8  10  9

[56]: A * B

[56]: 3x3 Matrix{Int64}:
        9   9   9
       12  12  12
       24  24  24
```

Рис. 15: Примеры работы с операциями над матрицами

Вывод

- В ходе выполнения лабораторной работы были получены навыки по подготовке рабочего пространства и инструментария для работы с языком программирования Julia, а также познакомились на простейших примерах с основами синтаксиса Julia

Список литературы. Библиография

[1] Julia Documentation: https://docs.julialang.org/en/v1/